

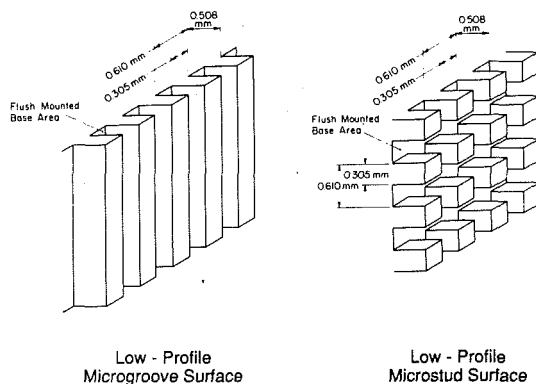


Fig. 3 Reference boiling curves used in the numerical effectiveness plots

to that of an isothermal surface at the same temperature regardless of surface orientation. The specific relationships between the heat flux and surface superheat were obtained from multi-segment linearized curve fits to data obtained for boiling on a flat vapor-blasted surface, and low-profile microgroove and microstud reference surfaces, Fig. 3, measured previously by the authors (Anderson and Mudawar, 1989). Since the incipience temperature excursion is surface specific, and practically nonexistent with extended cylindrical surfaces (as shall be demonstrated in the experimental results of the present study), the linearized boiling curves of the low-profile microgroove and microstud reference surfaces were adjusted in the incipience excursion region by a downward extrapolation of the linearized segment corresponding to nucleate boiling following incipience, to the point of intersection with the linearized fit to the natural convection data. Tables 1 and 2 give the power law coefficients for the linearized boiling curve data in the form

$$q = a(\Delta T_{\text{sat}})^n \quad (2)$$

where the power law coefficients a and n are valid only within the temperature range specified. The upper limits on wall su-

Table 1(a) Power law coefficients for the linearized smooth surface boiling curve (saturated FC-72)

$q(10^{-4} \text{ W/m}^2), \Delta T_{\text{sat}} (^{\circ}\text{C})$			
Region	a	n	ΔT_{sat} range
I	0.070305	0.961112	0.0–6.9
II	0.0042862	2.409421	6.9–9.2
III	6.20582×10^{-6}	5.355373	9.2–13.6
IV	0.043817	1.959946	13.6–17.6
V	1.771535	0.669954	17.6–38.1

Table 1(b) Power law coefficients for linearized smooth surface boiling curve (saturated FC-87)

$q(10^{-4} \text{ W/m}^2), \Delta T_{\text{sat}} (^{\circ}\text{C})$			
Region	a	n	ΔT_{sat} range
I	0.069711	1.072335	0.0–3.9
II	0.016067	2.150660	3.9–6.5
III	6.577022	0.025611	6.5–11.4
IV	0.088282	1.796897	11.4–15.1
V	2.596540	0.551379	15.1–33.5

Table 2(a) Power law coefficients for linearized low-profile microgroove reference surface boiling curve (saturated FC-72)

$q(10^{-4} \text{ W/m}^2), \Delta T_{\text{sat}} (^{\circ}\text{C})$			
Region	a	n	ΔT_{sat} range
I	0.066036	1.147633	0.0–5.1
II	1.874859×10^{-5}	6.166540	5.1–7.0
III	0.067283	1.960001	7.0–13.5
IV	0.181361	1.579018	13.5–21.0
V	5.080478	0.484374	21.0–35.3
VI	3.045850	0.627932	35.3–47.5

Table 2(b) Power law coefficients for linearized low-profile microstud reference surface boiling curve (saturated FC-72)

$q(10^{-4} \text{ W/m}^2), \Delta T_{\text{sat}} (^{\circ}\text{C})$			
Region	a	n	ΔT_{sat} range
I	0.112661	1.074969	0.0–4.7
II	1.587379×10^{-4}	5.296950	4.7–6.2
III	6.226659×10^{-3}	3.285856	6.2–8.9
IV	0.216195	1.663139	8.9–15.8
V	4.761849	0.541072	15.8–30.2
VI	2.865880	0.690070	30.2–45.6

perheat values in Tables 1 and 2 correspond to the CHF point associated with the individual reference surface.

Natural convection is sensitive to both surface geometry and orientation. To systematically model natural convection, the non-boiling regions of the experimentally-determined, linearized boiling curves were replaced with appropriate information based on geometry-specific correlations. The natural convection heat transfer from the perimeter of the cylinder was calculated by the Churchill and Chu (1975a) correlation for a horizontal cylinder,

$$\overline{Nu}_D = \left\{ 0.60 + \frac{0.387 Ra_D^{1/6}}{[1 + (0.559/Pr)^{9/16}]^{8/27}} \right\}^2 \quad (3)$$

The natural convection heat transfer coefficient for the circular tip was evaluated by dividing the tip area into thin vertical flat plates and integrating over the tip area. The heat transfer coefficient for each flat plate of length l was determined by Churchill and Chu's (1975b) correlation

$$\overline{Nu}_l = \left\{ 0.825 + \frac{0.387 Ra_l^{1/6}}{[1 + (0.492/Pr)^{9/16}]^{8/27}} \right\}^2 \quad (4)$$